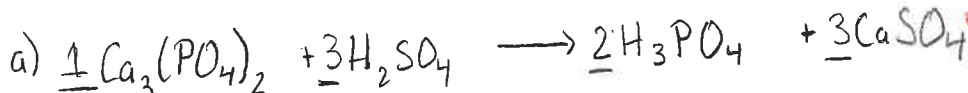


To Receive Full Credit, You Must Show All Your Work.
Be sure to use units correctly in all of your calculations.

Phosphoric acid (H_3PO_4) can be produced by reacting calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$) with a stoichiometric quantity of sulfuric acid (H_2SO_4). Calcium sulfate (CaSO_4) is also a byproduct of this reaction.

- (a) Write out the balanced chemical reaction equation. Make all of the stoichiometric coefficients the smallest whole numbers possible. Use any method you want to balance the reaction equation.
- (b) For 100 kg of H_2SO_4 , determine the kg of calcium phosphate consumed and the kg of phosphoric acid and calcium sulfate produced.
- (c) If calcium phosphate costs \$950/ metric-ton and sulfuric acid costs \$113/ metric-ton, and if phosphoric acid sells for \$960/ metric-ton and calcium sulfate sells for \$385/ metric-ton, what is the net profit or loss rate (\$/day) for the process based on selling 250 metric-ton/day of phosphoric acid produced only? Recalculate the profit/loss rate for this process considering that calcium sulfate can also be sold. (1 metric-ton = 1000 kg)

Species	Molar Mass
$\text{Ca}_3(\text{PO}_4)_2$	310.18
H_2SO_4	98.08
H_3PO_4	97.994 →
CaSO_4	136.14



Ca: 3

PO₄: 2

H: 6

SO₄: 3

Ca: 3

PO₄: 2

H: 6

SO₄: 3

b)

$$100 \text{ kg H}_2\text{SO}_4 \times \frac{1 \text{ mol H}_2\text{SO}_4}{0.09808 \text{ kg H}_2\text{SO}_4} \times \frac{1 \text{ mol Ca}_3(\text{PO}_4)_2}{3 \text{ mol H}_2\text{SO}_4} \times \frac{0.31018 \text{ kg Ca}_3(\text{PO}_4)_2}{1 \text{ mol Ca}_3(\text{PO}_4)_2} = 105.42 \text{ kg Ca}_3(\text{PO}_4)_2$$

$$100 \text{ kg H}_2\text{SO}_4 \times \frac{1 \text{ mol H}_2\text{SO}_4}{0.09808 \text{ kg H}_2\text{SO}_4} \times \frac{2 \text{ mol H}_3\text{PO}_4}{3 \text{ mol H}_2\text{SO}_4} \times \frac{0.097994 \text{ kg H}_3\text{PO}_4}{1 \text{ mol H}_3\text{PO}_4} = 66.61 \text{ kg H}_3\text{PO}_4$$

$$100 \text{ kg H}_2\text{SO}_4 \times \frac{1 \text{ mol H}_2\text{SO}_4}{0.09808 \text{ kg H}_2\text{SO}_4} \times \frac{3 \text{ mol CaSO}_4}{3 \text{ mol H}_2\text{SO}_4} \times \frac{0.13614 \text{ kg CaSO}_4}{1 \text{ mol CaSO}_4} = 138.81 \text{ kg CaSO}_4$$

c)

$$\text{Revenue} \left[\frac{250 \text{ t}}{\text{day}} \times \frac{\$960}{\text{t phosphoric}} \right] = \$240,000/\text{day} \quad P = R - C$$

$$\frac{250 \text{ t}}{\text{day}} \times \frac{\$385}{\text{t calcium sulfate}} = \$96,250/\text{day}$$

$$\frac{250 \text{ t}}{\text{day}} \times \frac{\$950}{\text{t calcium phosphate}} = \$237,500/\text{day}$$

$$\frac{250 \text{ t}}{\text{day}} \times \frac{\$113}{\text{t sulfuric acid}} = \$28,250/\text{day}$$

$$P = (\$240,000 + \$96,250) - (\$237,500 + \$28,250)$$

$$P = \$70,500/\text{day} \quad \text{profit if calcium sulfate is also produced}$$

$$P = (\$240,000) - (\$237,500 + \$28,250)$$

$$P = -\$25,750/\text{day} \quad \text{loss if only phosphoric acid produced}$$

